Lesson 5: Stretch and Release

Unit

Energy: Stores and Transfers

Lesson focus

Students use the spring constant from Lesson 4 to calculate the energy in the elastic potential store with $E_e = ½ k e^2$, and explain why doubling the extension gives four times the energy.

Specification

AQA GCSE Physics 4.1.1.2 Changes in energy and 4.5.3 Forces and elasticity. The equation is on the AQA Physics equation sheet.

Scope

Students calculate E_e, including when they first need k from F = k e. They are **not** expected to find e from a given E_e (it needs a square root).

Learning objectives

By the end of the lesson, students should be able to:

1. State that the work done in stretching a spring is stored in its elastic potential store.
2. Calculate elastic potential energy using $E_e = ½ k e^2$, with extension in metres.
3. Find k from a force and an extension, then use it to calculate E_e.
4. Explain why the equation only works up to the limit of proportionality.
5. Explain why doubling the extension gives four times the energy.

Key vocabulary

Keyword	Student-friendly definition
Elastic potential store	The energy an object has because it is stretched or squashed
Work done	Energy transferred when a force moves something
Compression	How much shorter a spring is when squashed; used in place of e

The equation

Quantity	Symbol	Unit
Elastic potential energy	E_e	joule (J)
Spring constant	k	newton per metre (N/m)
Extension	e	metre (m)

$$E_e = ½ k e^2$$

elastic potential energy = 0.5 × spring constant × extension²

It also works for a squashed spring, with e as the compression.

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Equipment and resources

- Each group's spring constant and graph from Lesson 4
- A spring-loaded toy (pop-up toy, jumping frog or toy launcher) for the starter
- Elastic bands, a toy car and a metre rule for the optional activity
- Calculators

Starter: Where did the jump come from?

Press down a pop-up toy and let it go. It leaps up.

Ask: which store was filled when I pushed it down? Where did that energy come from? (My muscles did work on it.) Where does it go? (Kinetic, then gravitational potential.)

> *The work done in stretching or squashing a spring is stored in its elastic potential store, as long as the spring is not permanently deformed.*

Main teaching: Calculating E_e

Worked example 1:

> *A spring with k = 50 N/m is stretched by 0.040 m. Calculate the energy in its elastic potential store.*
> $E_e = ½ k e^2$
> $E_e = 0.5 \times 50 \times 0.040^2$
> $E_e = 0.5 \times 50 \times 0.0016$
> $E_e = 0.04$ J

Point out how small this is. Lab springs store fractions of a joule; a bow or a bungee cord stores far more.

Worked example 2 (extension in cm, and a two-step):

> *A spring stretches by 10 cm when a 3.0 N force acts on it. Calculate the energy in its elastic potential store.*
> $e = 10$ cm $= 0.10$ m
> $F = k e$, so $k = F \div e$
> $k = 3.0 \div 0.10 = 30$ N/m
> $E_e = ½ k e^2$
> $E_e = 0.5 \times 30 \times 0.10^2$
> $E_e = 0.15$ J

Main teaching: Double the extension, four times the energy

Using the spring from example 1, stretch it to 0.080 m:

> $E_e = 0.5 \times 50 \times 0.080^2 = 0.16$ J

0.16 ÷ 0.04 = 4. Same reason as in Lesson 2: the extension is squared. Ask students what other equation in the unit has this pattern ($E_k = ½ m v^2$).

Why it is squared, for students who ask: the further you stretch the spring, the harder you have to pull. The second centimetre takes more force than the first, so more work is done per centimetre as it stretches.

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Main teaching: Only up to the limit of proportionality

Show the Lesson 4 graph again. The equation assumes the straight-line part, where k is constant. Beyond the limit of proportionality k changes, and some of the work done is not stored: if the spring is permanently stretched, it will not give all the energy back.

Activity 1: Your own spring

Each group uses its own k from Lesson 4:

1. Calculate E_e when the spring is stretched by 0.10 m, then 0.20 m.
2. Check that doubling the extension gave four times the energy.
3. Could you use the equation for the 700 g reading? Explain.

Sample answers with k = 25 N/m: 0.125 J and 0.50 J; no, it was beyond the limit of proportionality.

Activity 2: Calculation page

Workbook page: straight substitution; extensions in cm and mm; two-step questions (F and e given, find k, then E_e); compressed springs (a car's suspension) and stretched springs (a trampoline).

Activity 3: Elastic-band launcher (optional)

Groups launch a toy car with an elastic band stretched by 5 cm, 10 cm and 15 cm, and measure how far it travels. Discuss whether doubling the stretch doubles the distance, and why it might not (friction, the band is not a perfect spring). Keep this qualitative.

Safety: aim along the floor, never at people; wear goggles.

Check for understanding

1. What happens to the work done in stretching a spring?
2. Calculate E_e for a 200 N/m spring compressed by 0.05 m. (0.25 J)
3. A spring's extension is tripled. What happens to the energy stored? (Nine times as much.)
4. Why must the extension be in metres?

Plenary

Link the three stores so far. Students fill in:

Store	Equation	Depends on
Kinetic	$E_k = ½ m v^2$	mass, speed (squared)
Gravitational potential	$E_p = m g h$	mass, g, height
Elastic potential	$E_e = ½ k e^2$	spring constant, extension (squared)

Possible prep

Find a real object that uses an elastic potential store (a trampoline, a bow, a car's suspension, a mattress). Describe the energy transfers when it is used.

Teacher notes

1. Extension in centimetres is the commonest error, and it is a big one: 10 cm used as 10 instead of 0.10 gives an answer 10 000 times too large. Make students convert on a separate line.
2. Students often square k as well as e. Same fix as Lesson 2: write e^2 on its own line.

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3 Students do not need to know that E_e is the area under the force–extension graph.